

WG01: Kernel C++

Amlal El Mahrouss
NeKernel.org

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1 Introduction.

Many Operating Systems Kernels have been shipped using the C programming language.

And some of them like EKA2 uses the C++ programming language. Although notoriously difficult, one may still adapt to those constraints in order to deliver one such operating system kernel.

That is the reason that most production-grade kernels (Linux, XNU, and NT) are mostly written in C. With a higher-level subset in C++.

However, when correctly applying C++ principles to kernel development, one makes the development much more easier to pull off.

2 The Three Principles of Kernel C++.

2.1 Part One: The C++ Runtime

The problem mostly lies in the C++ runtime. Which assumes an existing host. A host is the contrary of a freestanding target, that is a program which expects a runtime to be present and linked to the program.

A C++ Kernel may instead make use of compile-time features of C++ alongside a tiny C++ runtime to make sure that no issues arise because of this host/freestanding difference.

2.2 Part Two: Constexpr and Friends.

One may avoid V-tables or runtime when possible. While focusing instead on meta-programming and compile-time features offered by C++. For example one may use templates to implement a scheduling policy algorithm. One example of such implementation may be:

```
1 /// Reference Implementation: https://github.com  
2 /// /nekernel-org/nekernel/blob/stable/src/kernel/  
3 /// KernelKit/CoreProcessScheduler.h#L51-L76
```

```

4  /// AND: https://github.com/nekernel-org/nekernel/blob/stable/
   src/kernel
5  /// /KernelKit/CoreProcessScheduler.h#L78-L105
6
7  struct FileTree final {
8      static constexpr bool is_virtual_memory = false;
9      static constexpr bool is_memory = false;
10     static constexpr bool is_file = true;
11     /// ...
12 };
13
14 struct MemoryTree final {
15     static constexpr bool is_virtual_memory = false;
16     static constexpr bool is_memory = true;
17     static constexpr bool is_file = false;
18     /// ...
19 };

```

As you see, two structures leverages the ‘constexpr’ keyword to make sure no bugs or panic occur at runtime because of a misuse of a system resource.

Which is why the constexpr keyword is very powerful here, we avoid the many pitfalls of writing (and hoping) that the C version will be well-thought enough so that we can catch such bugs later.

2.3 Bonus: Using Concepts and the DDK.

This bonus subsection intends to show how properly applied C++ Can solve issues related to driver validation. The following example shows how powerful C++ can be when combining it with Device Driver development as well.

```

1  /// Reference implementation:
2  /// https://github.com/nekernel-org/nekernel/blob/stable/src/libDDK/DriverKit
3  /// /c%2B%2B/driver_base.h
4  /// @brief This concept requires the driver
5  /// to be IDriverBase compliant.
6
7
8  inline constexpr auto kInvalidType = 0;
9
10 template <class Driver>
11 concept IsValidDriver = requires(Driver drv) {
12     { drv.IsActive() && drv.Type() > kInvalidType };
13 };
14
15 /// @brief Consteval helper to detect
16 /// whether a template is truly based on IDriverBase.
17 /// @note This helper is consteval only.
18 template<IsValidDriver T>
19 consteval void ce_ddk_is_valid(T) {}

```

2.4 Part Three: Memory and C++ Classes.

This last part treats about the final and most important part of this paper so far. Memory. As you may already have known, the C++ language uses a class lookup system (also called v-table) in order to refer to a base method in case if the instance has it missing.

```
1  /// Link: https://godbolt.org/z/q3Wceannr
2  /// /std:c++20 /Wall
3
4  #include <iostream>
5
6  class A
7  {
8  public:
9      explicit A() = default;
10     virtual ~A() = default;
11
12     virtual void doImpl()
13     {
14         std::cout << "doImpl()\r\n";
15     }
16 };
17
18 class B : public A
19 {
20 public:
21     explicit B() = default;
22     ~B() override = default;
23 };
24
25 int main() {
26     B callImpl;
27     callImpl.doImpl();
28 }
```

3 Addressing V-Tables in a C++ Kernel.

Now the problem with kernel development is that we want to avoid such feature as much as possible, and we'd do that by following the Prong on Inheritance:

3.1 The Three Prongs on Inheritance:

TTPI is a thought exercise used to decide whether you should consider using C++ in a kernel. Consider the following thought exercise:

- 1: Is this a feature that can be implemented with other similar protocols/concepts?

2: Is this doable without too much trade-off costs?

3: Is this doable without V-Tables?

If 2/3 of those questions fail, you should consider finding another solution to your problem. As it surely has an equivalent without the problematic points.

4 Conclusion

We can now conclude that C++ in Kernel Development is indeed possible under a subset of C++, here called Kernel C++.

A reference implementation of this paper exists, It's called NeKernel.org, available under the same internet address.

I am looking forward to any questions or inquiries at: amlal@nekernel.org.

4.1 Additional Resources

EKA2: wikipedia.org

XNU: wikipedia.org

NeKernel.org: <https://nekernel.org>

5 References

NeKernel.org: NeKernel.org, <https://nekernel.org>

Sales. J, Tasker. M: Introducing EKA2, <https://media.wiley.com>

Driesen. K, Hölzle. U: The direct cost of virtual function calls in C++, <https://dl.acm.org>